Thursday
19/03/26

(1) Consider a simple paging system with 8 KB page size and a page Table with each entry of size 4 bytes. Answer the following questions

(a) How many bits are used for representing the page offset value?

(b) What is the size of the physical memory in bytes that can be addressed?

(c.) Calculate the amount of Internal fragmentation for a process of size 205 KB?

(d) Is it possible to load a process of size 98 KB if there are 12 free frames? Justify your Answer.

Ans) Given,

page size = 8 KB = 8 × 1024 = 8192 bytes

page Table entry size = 4 bytes.

1024
× 8
8192

(a.) Number of Bits for page offset :

page offset bits depends on page size,

Page size = 8192 bytes

Number of bits for page offset = $\log_2(8192)$

$\Rightarrow \log_2(2^{13}) = 13$ bits.

(b.) Size of physical memory that can be addressed:

page Table Entry = 4 bytes = 4 × 8 = 32 bits

Out of this 32 bits, Offset = 13 bits

Remaining bits = frame number bits.

So, frame number bits = 32 − 13 = 19 bits

Number of frames = 2^{19} frames

Each frame = 8 KB = 2^{13}

∴ Total physical Memory = $2^{19} \times 2^{13}$

$\Rightarrow 2^{32}$ bytes

(c) Internal fragmentation for process size 205 KB,

Page size = 8 KB

Number of pages required for the process $= \frac{205}{8}$

$= 25.6 \Rightarrow 26$ pages

Total allocated memory for the process = 208 KB

$$\begin{array}{r} 26 \\ \times\ 8 \\ \hline 208 \end{array}$$

Internal fragmentation = 208 − 205

$\Rightarrow$ 3 KB

(d.) Page size / frame size = 8 KB

Total pages require for the process $= \frac{98}{8}$

$= 12.25 = 13$ frames

$$\begin{array}{r} 12.25 \\ 8\overline{)98} \\ 8 \\ \hline 18 \\ 16 \\ \hline 20 \\ 16 \\ \hline 40 \end{array}$$

And the available frames = 12

∴ Not possible to load the process.

* Page Table Size Calculation :-

eg: A 32 bit system with 4 KB pages (2^{12} bytes) and a 4 byte page Table Entry Find the page Table size?

Ans) Total Number of Entries $= \frac{2^{32}}{2^{12}} = \underline{\underline{2^{20}}}$

Page Table Entry $= 4$ bytes

$\therefore$ Page Table size = Total Entries × size of single page table entry

$= 2^{20} \times 4$ bytes

$= 2^{20} \times 2^{2} = \underline{\underline{2^{22}}}$ bytes.